



— ENGINEERING & ROBOTICS · CAPSTONE 2026

Navigation & Guidance of Blind and Visually-Impaired People

Camera-free swarm robotics and a simple app that give blind and visually impaired people the freedom to move independently, safely, and on their own terms — guided by two or more robots following and leading them.

Presented by **Violet** · **Samuel** · **Gabriel** · July 2026

Independent indoor travel is still out of reach.

2.2B

people live with a vision impairment; blind people rise from ~43M (2020) to ~61M by 2050.

28%

of blind & partially-sighted people feel able to travel independently — even after training.

Indoors

no curbs, walls or traffic sounds as landmarks — an unfamiliar building is "too hard to attempt."

\$45–60k

a guide dog: 1–5 years to train, serves only ~2% of blind people (<19,000 working dogs, falling).

40%+

of white-cane users report head-level accidents monthly — the cane senses only below the waist.

Audio-only

BlindSquare, GoodMaps, Evelity, Wayfindr — every product talks; **none physically leads.**

PROBLEM STATEMENT

How might we help a blind person navigate an unfamiliar indoor space safely and independently — without a guide dog, a costly wait, or relying on another person?

Three low-cost robots that physically lead the way.

A system of three small robots plus a phone app that leads a blind traveler (Sam) along pre-mapped indoor routes. Two guide robots lead, a companion stays beside Sam, and one piece of software — the brain — coordinates all three over Bluetooth. **No camera. No internet.**

ASK & PLAN

- **Voice destination** — "Take me to the store."
- **Route + ETA** — "be there by 6:00 PM" and it departs on time by itself.

COORDINATE

- **Real-time (~5 Hz)** — balances speed & spacing so no one is left behind.
- **Safety** — stops on obstacles, taps the arm, speaks every step; emergency stop anytime.

PRIVATE & SHARED

- **Camera-free & offline** — private by design (hospitals, care homes).
- **Live family view** — a relative anywhere can follow the journey.

What it must do — and why it matters.

OBJECTIVES

- Lead a blind user from A to B along a mapped route, **reliably**.
- Hold a safe formation — the slowest member (Sam) **sets the pace**.
- Stop at obstacles and alert the user — **light + arm tap + voice**.
- Compute ETA and support **scheduled departures**.
- Stay **low-cost, camera-free and offline**.
- Provide **live monitoring** for family and caregivers.

EXPECTED IMPACT

- Restores **independent indoor mobility** for a large, ageing, under-served population.
- A kit could target **under \$2,000** — vs a \$45–60k, multi-year guide dog.
- **Physically guides** the user — unlike every audio-only app.
- Fits accessibility mandates (**EAA, ADA**) — airports, care homes, museums, hospitals.
- Assistive-independence programs return **\$4–7 of social value per \$1 (SROI)** — plus dignity, employment, mental health.

Simple robots. One intelligent brain.

HARDWARE

- **3× XRP robots** (RP2350, MicroPython) — driven wheels + encoders, IMU, two line sensors, HC-SR04 ultrasonic, servo "arm", LED.
- **Roles:** Opener (sensors) · Sam (blind, no external sensors) · Closer (follower).
- **The hub** — a phone or laptop running the app.
- **Bluetooth** (Nordic UART) links all three — no internet, ~\$2,000 target kit.

SOFTWARE

- **One identical firmware** on all three — a role-based state machine: line-follow, marker & dead-reckoning navigation, gap-holding.
- **The hub app:** city map + route planning + ETA, ~5 Hz coordination, voice interface.
- **Reliable command protocol** — acknowledged, retried, verified.
- Camera-free and fully **offline**.

From three kits to a coordinated convoy.

BUILDING PROCESS

- Assembled **3 XRP kits**; wrote **one firmware** shared by all — roles assigned live over Bluetooth.
- Built a web-based **Hub** (Web Bluetooth) that connects the fleet, plans routes on a hand-mapped model town, and speaks each step.
- Added a **voice interface** — spoken destination in, spoken guidance out.

INITIAL TESTING

- Verified **BLE connection + role assignment** and the live telemetry loop.
- Confirmed **motor control** and per-robot state reporting.
- Built a full **software simulator** of the 3-robot convoy to validate the coordination logic end-to-end **before touching hardware**.

The real challenge was wireless reliability — not the motors.

ENVIRONMENTS

- A hand-built **miniature town**, measured to the centimetre.
- A full **simulator** (virtual fleet) with **injectable packet loss**.

FINDINGS

- Commands could be **lost, delayed, or duplicated** — a robot left behind.
- Sensors were noisy — **ultrasonic spikes, encoder glitches**.

ITERATIONS

- Rebuilt comms: **acknowledgment, retry, synchronized start, verification**.
- An **adversarial review** found & fixed **10 reliability defects**.
- Hardened firmware — **idempotent commands, debounced obstacle detection**.

✓ Delivery holds at 40% simulated packet loss → still 3/3

A swarm you can trust to move together.

SUCCESS METRICS

- Commands **confirmed by all 3** — even at 40% simulated packet loss.
- **Deterministic**: the fleet starts in sync, or safely aborts.
- Full **voice → coordinate → guide → arrive** validated in simulation.

USER-CENTRED DESIGN

- Camera-free & offline — answers the **privacy** concern head-on.
- The user **sets the pace** — the system adapts to them, not the reverse.
- Spoken guidance + arm tap for a **non-visual** experience.

CHALLENGES OVERCOME

- **BLE reliability** — a verifiable, retrying protocol.
- **Sensor noise** — filtering & debouncing.
- **Synchronization** — a two-phase, verified start.

Where it goes next.

SENSE

- **Head-height obstacle detection** — the hazard the cane misses.
- **Haptic feedback** in the companion robot.

SCALE

- **Larger swarms** and longer routes.
- **Automatic mapping** of new indoor spaces.
- Better **battery & range**.

DEPLOY

- **Real-world pilots** — airports, care homes, hospitals.
- **Trials with the people we built it for.**

Every step toward the same goal: guidance that gives independence back.

Problem identified. Solution built. Coordination proven.

SUMMARY

- **Problem** — a large, under-served population can't move independently indoors.
- **Solution** — a camera-free, voice-first robot swarm that physically guides.
- **Result** — a reliable, verifiable coordination layer, validated end-to-end.

LIVE DEMONSTRATION

- Speak a destination — "**Take me to the Eiffel Tower.**"
- The hub confirms all robots, then triggers a **synchronized start**.
- The fleet moves through the town and **announces arrival**.

Q & A

Open floor — we'd love your questions.



— THANK YOU

Thank you.

To our mentors, teachers, and everyone who tested, questioned, and cheered us on — thank you. This project is for the travelers it's meant to serve.

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